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CSC 34200

HW 3 Creating Rational Number Program

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Objective: We are tasked to create a program that create two rational number that can do arithmetic operations.

C++ Code:

```
1  *****
2  Allan Yunayev
3  hw make a rational program that add, subtracts, multiply and divide
4  *****
5  #include <iostream>
6  using namespace std;
7  int main()
8  {
9      int P,Q,M,N, testcase;
10     cout<<" Enter your first rational number"<<endl;
11     cin>>P>>Q;
12     cout<<" Enter your second rational number"<<endl;
13     cin>>M>>N;
14
15     while (N == 0 || Q ==0){
16         if(N == 0){
17             cout<< "Your N can not be 0 enter another number"<<endl;
18             cin>>N;
19         }
20         if(Q == 0){
21             cout<< "Your Q can not be 0 enter another number"<<endl;
22             cin>>Q;
23         }
24     }// to make sure the second pair of rational is not = to 0
25
26     cout<<"\nPress 1: Add 2: Subtract 3: Multiply 4: Divide\n";
27     cin>>testcase;
28     cout<<endl;
29
30     switch(testcase){
31         case 1:
32             int addNum;
33             int addDem;
34             if(Q*N ==0){
35                 addNum=(P)+(M*(Q/N));
36                 addDem=(Q);
37             }
38             else if(N*Q ==0){
39                 addNum=(P*(N/Q))+(M);
40                 addDem=(N);
41             }
42
43             else{
44                 addNum= (P*N)+(M*Q);
```

```

45     addDem= (N*Q);
46 }
47 cout<<"The addition of ( "<<P<<" , "<<Q<<" ) and ( "<<M<<" , "<<N<<" ) is ( "<<addNum<<" , "<<addDem<<" )\n";
48 break;
49
50 case 2:
51     int subNum;
52     int subDem;
53     if(Q%N ==0){
54         subNum=(P)-(M*(Q/N));
55         subDem=(Q);
56     }
57     else if(N%Q ==0){
58         subNum=(P*(N/Q))-(M);
59         subDem=(N);
60     }
61
62     else{
63         subNum= (P*N)-(M*Q);
64         subDem= (N*Q);
65     }
66     cout<<"The subtraction of ( "<<P<<" , "<<Q<<" ) and ( "<<M<<" , "<<N<<" ) is ( "<<subNum<<" , "<<subDem<<" )\n";
67     break;
68
69 case 3:{
70
71     int multNum = (P*M);
72     int multDem = (Q*N);
73     cout<<"The Multiplication of ( "<<P<<" , "<<Q<<" ) and ( "<<M<<" , "<<N<<" ) is ( "<<multNum<<" , "<<multDem<<" )\n";
74     break;
75 }// for some reason it give an error if i dont include the {}; |
76 case 4:
77     int divNum=(P*N);
78     int divDem=(Q*M);
79     cout<<"The division of ( "<<P<<" , "<<Q<<" ) and ( "<<M<<" , "<<N<<" ) is ( "<<divNum<<" , "<<divDem<<" )\n";
80     break;
81 }
82
83
84 return 0;
85 }

```

Addition Case 1:

```

Enter your first rational number
2
4
Enter your second rational number
1
8

Press 1: Add 2: Subtract 3: Multiply 4: Divide
1

The addition of ( 2 , 4 ) and ( 1 , 8 ) is ( 5 , 8 )

...Program finished with exit code 0
Press ENTER to exit console.

```

Subtraction Case 2:

```
Enter your first rational number
2
4
Enter your second rational number
1
8

Press 1: Add 2: Subtract 3: Multiply 4: Divide
2

The subtraction of ( 2 , 4 ) and ( 1 , 8 ) is ( 3 , 8 )

...Program finished with exit code 0
Press ENTER to exit console.
```

Multiplication Case 3:

```
Enter your first rational number
2
4
Enter your second rational number
1
8

Press 1: Add 2: Subtract 3: Multiply 4: Divide
3

The Multiplication of ( 2 , 4 ) and ( 1 , 8 ) is ( 2 , 32 )

...Program finished with exit code 0
Press ENTER to exit console.
```

Division Case 4:

```
Enter your first rational number
2
4
Enter your second rational number
1
8

Press 1: Add 2: Subtract 3: Multiply 4: Divide
4

The division of ( 2 , 4 ) and ( 1 , 8 ) is ( 16 , 4 )

...Program finished with exit code 0
Press ENTER to exit console.
```

Explanation: What I did in this code the allow the user to create their own rational numbers but if the denominator is 0 then it will go in a loop until the denominator is changed to a non-zero number.

After that I used a switch case to do add, subtraction, multiplying or dividing. For adding the rational numbers if the one of the denominators is a multiple of the other than we will adjusted the rational number to have the denominator with the lowest common denominator.

For subtraction it the same thing but we are subtracting not adding.

For multiplying we multiply the numerators together and the denominators together to get our result

Lastly for dividing we flip the numerator and denominator of the one of the rational number and multiply them as if we were doing multiplication.